

# Ai Assisted Coding

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## Responsible AI

### Task Description #1 (Privacy& Security in File Handling)

**Task:** Use an AI tool to generate a Python script that stores user data (name, email, password) in a file.

**Analyze:** Check if the AI stores sensitive data in plain text or without encryption.

### Input

```
1  import hashlib
2
3  name = input("Enter name: ")
4  email = input("Enter email: ")
5  password = input("Enter password: ")
6
7  hashed_password = hashlib.sha256(
8      password.encode()
9  ).hexdigest()
10
11 with open("user_data.txt", "w") as file:
12     file.write(f"Name: {name}\n")
13     file.write(f"Email: {email}\n")
14     file.write(f>Password Hash: {hashed_password}\n")
15
16 print("User data saved securely.")
17 print("Password is stored as a hash.")
```

## Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task1.py

Enter name: Lakshitha
Enter email: lakshitha@gmail.com
Enter password: abc123

User data saved securely.
Password is stored as a hash.
```

## Task Description #2– Fair Decision Logic

**Task:** Prompt an AI tool to generate a scholarship eligibility checker based on academic score, family income, and location. Examine whether the logic unfairly favors or disadvantages any group.

## Input

```

1  academic_score = float(input("Enter academic score: "))
2  income = float(input("Enter family income: "))
3  location = input("Enter location: ")
4
5  eligible = (
6      academic_score >= 75
7      and income <= 300000
8      and location.lower() in ["rural", "urban"]
9  )
10
11 if eligible:
12     print("Scholarship Eligible")
13 else:
14     print("Scholarship Not Eligible")
15
16 print("Decision is based on score and income.")
17 print("Location is not used to favor a group.")

```

## Output

```

PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task2.py

Enter academic score: 82
Enter family income: 250000
Enter location: rural

Scholarship Eligible
Decision is based on score and income.
Location is not used to favor a group.

```

## Task Description #3 (Transparency in AI-Based Admission System)

**Task:** Use an AI tool to generate a Python program that predicts whether a student should be selected or rejected for admission based on factors such as entrance exam score, academic marks, and other relevant admission criteria.

**Analyze:** Check whether the AI system provides a clear and understandable reason for its admission decision.

## Input

```
1  entrance = float(input("Enter entrance exam score: "))
2  marks = float(input("Enter academic marks: "))
3
4  score = (entrance * 0.6) + (marks * 0.4)
5
6  if score >= 70:
7      decision = "Selected"
8      reason = "The combined score meets the admission threshold."
9  else:
10     decision = "Rejected"
11     reason = "The combined score is below the admission threshold."
12
13  print("Admission Decision:", decision)
14  print("Reason:", reason)
15  print("Combined Score:", round(score, 2))
```

## Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task3.py

Enter entrance exam score: 78
Enter academic marks: 82

Admission Decision: Selected
Reason: The combined score meets the admission threshold.
Combined Score: 79.6
```

## Task Description #4 – Ethical Scoring System

**Task:** Generate an employee performance evaluation system using inputs such as project completion rate, teamwork score, and

attendance. Analyze the scoring logic for any unethical weighting or hidden bias.

## Input

```
1  completion = float(input("Project completion rate (%): "))
2  teamwork = float(input("Teamwork score (%): "))
3  attendance = float(input("Attendance (%): "))
4
5  performance = (
6      completion * 0.50
7      + teamwork * 0.30
8      + attendance * 0.20
9  )
10
11 print("Performance Score:", round(performance, 2))
12
13 if performance >= 75:
14     print("Evaluation: Good Performance")
15 else:
16     print("Evaluation: Needs Improvement")
17
18 print("Weights are based on job-related criteria.")
```

## Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task4.py

Project completion rate (%): 90
Teamwork score (%): 80
Attendance (%): 95

Performance Score: 87.0
Evaluation: Good Performance
Weights are based on job-related criteria.
```

## Task Description #5 – Accessibility and Inclusiveness

**Task:** Use an AI tool to generate a user feedback form application. Review whether the language and options provided are inclusive and accessible to diverse users.

## Input

```
1  print("=== User Feedback Form ===")
2
3  name = input("Name (optional): ")
4  feedback = input("Please share your feedback: ")
5
6  print("\nHow was your experience?")
7  print("1. Very Good")
8  print("2. Good")
9  print("3. Neutral")
10 print("4. Needs Improvement")
11 rating = input("Choose an option (1-4): ")
12
13 if rating in ["1", "2", "3", "4"]:
14     print("\nThank you for your feedback!")
15     print("Your response has been recorded.")
16 else:
17     print("\nPlease choose an option from 1 to 4.")
18
19 print("Name is optional and the form uses clear language.")
```

## Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task5.py

=== User Feedback Form ===
Name (optional): Lakshitha
Please share your feedback: Easy to use

How was your experience?
1. Very Good
2. Good
3. Neutral
4. Needs Improvement
Choose an option (1-4): 1

Thank you for your feedback!
Your response has been recorded.
Name is optional and the form uses clear language.
```

## Task Description #6 (Fairness in Loan Approval System)

**Task:** Use an AI tool to generate a Python program that predicts whether a person should be approved or rejected for a loan based on information such as income, age, employment status, credit score, and loan amount.

**Analyze:** Check whether the AI-generated system makes decisions unfairly based on gender, religion, race, or other sensitive/unrelated personal attributes.

## Input

```
1  income = float(input("Enter annual income: "))
2  age = int(input("Enter age: "))
3  employment = input("Employment status: ")
4  credit_score = int(input("Enter credit score: "))
5  loan_amount = float(input("Enter loan amount: "))
6
7  approved = (
8      income >= 300000
9      and age >= 18
10     and employment.lower() == "employed"
11     and credit_score >= 650
12     and loan_amount <= income * 0.5
13 )
14
15 if approved:
16     print("Loan Decision: Approved")
17     print("Reason: Financial criteria are satisfied.")
18 else:
19     print("Loan Decision: Rejected")
20     print("Reason: One or more financial criteria are not satisfied.")
21
22 print("No gender, religion, or race is used.")
```

## Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task6.py

Enter annual income: 600000
Enter age: 25
Employment status: employed
Enter credit score: 720
Enter loan amount: 200000

Loan Decision: Approved
Reason: Financial criteria are satisfied.
No gender, religion, or race is used.
```